

QRC / ARCS: Project Changes & Execution Plan

The engineering/repo side of the plan — your 10 requested changes, made concrete and sequenced

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Companion document	“QRC / ARCS: Research Plan & Paper Structure” — the writing/research side of this same plan

1. Hardware/Circuit Simulation Expansion

1.1 Cross-validate in NI Multisim

Rebuild the three existing LTSpice stages (power supply, LC resonator, envelope detector) in Multisim as an independent SPICE engine. Report the same headline numbers (rail voltage, resonant frequency, Lorentzian peak shape) from both tools side by side — tool-independent agreement is real, cheap credibility.

1.2 Frequency behavior beyond the nominal drift band

- Run an AC sweep well outside the ± 25 kHz drift band used by the RL environment (e.g., 100 kHz–2 MHz) to check for secondary resonances, parasitic effects, or op-amp gain-bandwidth roll-off that the narrow-band RL model doesn't see.
- Document the OPA2134UA's gain-bandwidth product against the actual signal frequencies in play — confirm it isn't the limiting factor at the top of the swept range.

1.3 Temperature behavior

- No physical hardware exists, so “high temperature” behavior means a SPICE temperature sweep (the .TEMP directive in LTSpice/Multisim), not a bench test. Sweep across a realistic operating range (e.g., -20°C to 85°C) and plot: (a) resonant-frequency shift vs. temperature, driven mainly by the varactor's and inductor's temperature coefficients; (b) the AMS1117-3.3 regulator's line/load and thermal regulation; (c) op-amp offset/bias drift. Pull the actual tempco figures from the BB909/MV209 (or BB135,115 substitute) and SRR1280-100M datasheets rather than assuming a number.

1.4 Justify every component choice explicitly, in writing

Component/decision	Justification to write up
C0G capacitors (not X7R) for the LP filter and varactor coupling	X7R's capacitance varies significantly with applied voltage and temperature; C0G is stable — state this quantitatively (typical X7R capacitance swing vs. C0G) rather than asserting it
3 cm maximum RF lead length	Tie to the wavelength/parasitic-inductance argument at $\sim 500\text{--}556$ kHz — show the calculation, not just the rule
Separate analog/digital AMS1117-3.3 rails	Justify via digital switching noise coupling into the analog resonator path if rails were shared
DAC/DDS choice for excitation (see §4 on reconciling MCP4921 vs. AD9834)	Once reconciled, state why the final choice was made (12-bit SPI DAC simplicity vs. DDS's cleaner spectral purity for a fixed-amplitude sine excitation)

2. Virtual Hardware-in-the-Loop: Dry Runs and Model Deployment

This is the closest legitimate substitute for physical validation available to you (established in the prior report) — now made into concrete steps:

Step	Action
1	Build a Simulink/Simscape Electrical model of the full signal chain: resonator response → envelope detector → RC filter → 12-bit ADC quantization → (reconciled DAC/DDS) excitation path, including realistic component non-idealities (op-amp GBW/slew limits, SPI transfer timing).

Step	Action
2	Dry run: feed the virtual plant a handful of fixed test frequencies and confirm the output amplitude curve matches the expected Lorentzian shape and the ADC codes are sane, before running any RL policy against it.
3	Deploy: run the frozen V4 policy (5 seeds) through the virtual-HIL plant for the same number of episodes used in the classical Gym evaluation.
4	Compare MAE/near-resonance/low-amplitude between the abstract Gym environment and the virtual-HIL plant — report both, labeled clearly as “idealized simulation” vs. “component-level simulated plant,” never as “hardware.”

3. No Hardcoded Results — Provenance Policy

File	Current problem	Fix
pid_baseline.py	Hardcodes RL numbers for comparison in its summary printout	Load RL results from the actual saved evaluation logs/JSON at run time
compare_v3_v4.py	Hardcodes V3/V4 numbers	Same fix — compute from saved checkpoints/logs, never paste literals
Every results JSON going forward	No record of how a number was produced	Add a small provenance block to every saved result: git commit hash, seed(s) used, timestamp, exact command run — so any number in the paper can be traced back to a regenerable run

Add one CI-style check if you can spare the time: a script that scans results files for the provenance block and fails loudly if it's missing. Cheap insurance against exactly the kind of hardcoding the agentic audit already caught once.

4. Path Migration and Pipeline Repair

Step	Action
1	grep the entire repo for absolute path fragments (C:\MiniProject_Sem4, C:\Projects\MiniProject_Sem4, any other machine-specific paths).
2	Replace every hit with a path resolved relative to the repo root (e.g., a single PROJECT_ROOT constant derived from the current file's location, or a small config/.env file read once at startup) — never a second hardcoded absolute path.
3	Restore or recreate qiskit_integration.qiskit_resonator_env_v2 (only _v3 currently exists), or repoint phase1_baseline.py and sweep_verify_v2.py at whichever version is now canonical.
4	Prove it: copy the whole repo to a fresh directory (simulating a new machine/collaborator) and run the full pipeline end-to-end from that copy before considering this done.

5. Plotting and Comparative Charts

This is genuinely one of the highest-value items on your list — a reviewer will look at your figures before your tables. Build these specifically:

Chart	Data source	Type	Purpose
Ablation comparison	V3 / V4a / V4b / V4 (5-seed aggregate)	Grouped bar chart, MAE with error bars	The paper's single most important figure — makes the negative-information finding immediately visible
Noise-regime comparison	Gaussian vs. OU, per controller (PID/V3/V4)	Grouped bar chart	Shows the regime-dependence of the RL advantage at a glance
PID variance story	PID Gaussian per-episode MAE distribution vs. V4's	Box plot or violin plot	Makes the “PID fails catastrophically on a subset of episodes” argument visual rather than just a std-dev number
Trajectory plots	Agent frequency vs. true resonance over an episode, for a representative success and a representative near-failure	Line plot, one or two panels	Your own established safeguard against degenerate policies — belongs in the paper as a figure, not just an internal check

Chart	Data source	Type	Purpose
Cross-simulator transfer	Classical / misaligned-Qiskit / aligned-Qiskit / fine-tuned-Qiskit	Grouped bar chart, MAE + near-resonance side by side	Tells the whole QACT story in one figure
Per-episode QACT variance breakdown	QACT episode-level MAE values	Histogram or scatter, seeded rerun	Shows whether a subset of episodes drives the large QACT variance, as recommended previously
Equipment/algorithm sensitivity (if §8 of the companion research document is run)	Sensitivity grid results	Heatmap	Visualizes the two-axis sensitivity story in one panel

6. Fix the Quantum Environment's Physics/Alignment Issues

Condensed checklist (full rationale and evidence already established in the prior planning document):

Fix	Detail
Align the observation vector	Match classical V4's exact order and scaling: [norm_freq, ema_amp, prev_ema_amp, prev_action, amp_gradient], gradient clipped to ± 1 , computed as $\Delta \text{amp} \cdot 3000 / (\Delta \text{freq} + 1)$.
Match or normalize the resonance linewidth	Raise omega_rabi to ≈ 5000 Hz to match the classical ± 3333 Hz near-resonance band, or keep 2500 Hz and report a width-normalized metric with the asymmetry disclosed.
Port the full shaped reward	Gradient bonus, action-cost penalty, low-amplitude penalty, 3-step dead termination — not reward = amp alone.
Restore action_scale = 3000	Currently cut to 1000 in the Qiskit env, making the agent under-agile relative to the drift rate.
Seed everything	AerSimulator, every RNG in the classical and Qiskit envs, and the PID grid search.

7. Zero-Shot Then Fine-Tuned: The Exact Quantum-Environment Protocol

Step	Action	Record
A	With all §6 fixes applied, run the current (frozen) V4 policy zero-shot in the aligned Qiskit environment, 5 seeds, ≥ 4096 shots per evaluation.	MAE mean $\pm$ std, near-resonance, low-amplitude — this is the “zero-shot cross-simulator transfer” number
B	Fine-tune from that same starting checkpoint for a small, explicitly-stated budget (e.g., 5–10% of the original 2M-step budget) inside the aligned Qiskit environment, with multiple parallel envs rather than one DummyVecEnv.	Save as a separately labeled checkpoint (e.g., V4-QACT-finetuned) — never overwrite or retrain the locked V4 itself
C	Evaluate the fine-tuned checkpoint the same way as Step A, same 5 seeds, same shot budget.	MAE mean $\pm$ std, near-resonance, low-amplitude — the “fine-tuned transfer” number
D	Report A and C side by side, plus the classical V4 aggregate for reference, in the §8 table below.	This full triplet is the actual publishable result — not either number alone

8. Classical-to-Quantum Tabulation Template

Environment	MAE (Hz)	Near-resonance	Low-amplitude	Seeds	Notes
Classical V4 (aggregate)	2205 $\pm$ 518	84.8% $\pm$ 7.6%	5.1% $\pm$ 3.7%	5	Reference / upper bound
Qiskit, as-committed (misaligned)	2964 $\pm$ 2473	32.8%	21.2%	1 (diagnostic only)	Kept only to show the effect of the alignment fix — not a result to build conclusions on
Qiskit, aligned, zero-shot	[fill in from §7 Step A]	[fill in]	[fill in]	5	The real zero-shot transfer number

Environment	MAE (Hz)	Near-resonance	Low-amplitude	Seeds	Notes
Qiskit, aligned, fine-tuned	[fill in from §7 Step C]	[fill in]	[fill in]	5	The real fine-tuned transfer number

9. On Reframing as a “Review Paper”

Covered in full, with the specific IEEE Access editorial guidance behind it, in the companion research document's §0. Short version: keep the manuscript type as “Research Article” and get everything you actually wanted (methodology depth, comparisons, future aspects) through a strong related-work/comparative section — declaring it a “Review” or “Survey” risks a desk rejection on type-mismatch grounds given how much original experimental data this paper carries.

10. Repository Restructure and Cleanup

10.1 Files to resolve, not just delete

File(s)	Issue	Recommended action
rl_environment_v4.py	Described as an identical backup copy of the real V4 env	Delete; rely on git history for the backup, not a duplicate file
train_v4_gaussian.py	Described as an earlier backup of train.py	Delete or move under a clearly labeled /archive folder excluded from the published repo
“rl_environment copy.py” (the V3 env)	Ambiguous, unprofessional filename for a file you actually need (it's the V3 baseline)	Rename to rl_environment_v3.py and keep — this is load-bearing, not clutter
phase1_baseline.py, sweep_verify_v2.py	Currently broken (missing import)	Fix per §4; only delete if you decide they're genuinely redundant with a working script
AGENTS.md, CLAUDE.md, .claude/settings.json, .opencode/ (incl. node_modules)	AI-tool orchestration configuration, not project science	Exclude from the published/paper-facing repo (e.g., via .gitignore or a separate tooling branch) — including these in a repo you share alongside a paper reads as clutter unrelated to the research

10.2 Proposed final structure

Folder	Contents
/simulation	QuantumResonatorSim, both drift models, clearly named (no “copy”/“backup” files)
/rl_training	One canonical file per variant (V3, V4, V4a, V4b, and the new confound-isolation variants from the research plan §10), train.py, evaluate_agent.py — all reading paths from the fix in §4
/qiskit_integration	Fixed, reproducible QACT pipeline only — no dead imports
/circuit_sim	LTSpice + Multisim + Simulink files together, clearly labeled by stage
/firmware	esp32_controller.ino, explicitly marked in a header comment as “designed, simulated, not deployed on physical hardware”
/results	Final tables/plots/JSON, each with the provenance block from §3
/paper	Manuscript drafts
README.md	One page: environment setup, and the exact command to regenerate every table and figure in the paper from a clean checkout

10.3 Before/after check

Do the cleanup only after §3–§8 fixes are complete and verified — restructuring a repo that still has broken imports and hardcoded paths just moves the same problems to new file locations. Sequence: fix → verify end-to-end from a clean checkout → then restructure and delete.

11. Consolidated Execution Timeline

Phase	What	From
0 — Integrity fixes	Report the 5-seed aggregate, not Seed 3 alone; stop citing unreproducible QACT numbers; fix dead imports/paths (§4); seed everything; de-hardcode comparison scripts (§3).	§3, §4
1 — Confound ablation	Run the two observation-vs-reward isolation variants, 5 seeds each (research document §10, Step 4).	Research doc §10
2 — Quantum environment rebuild	Apply all §6 alignment fixes; run the §7 zero-shot → fine-tune protocol; populate the §8 table.	§6, §7, §8
3 — Ablation finalization	Re-run V4a/V4b under both Gaussian and OU regimes, 5 seeds each, matching V4's protocol.	Research doc §5
4 — Circuit/front-end validation	Multisim cross-check, temperature/frequency sweeps, component justification write-up, virtual-HIL dry run and deployment (§1, §2).	§1, §2
5 — Sensitivity study (optional but recommended)	Two-axis equipment/algorithm sweep (research document §8).	Research doc §8
6 — Plotting	Build the full chart set in §5 once all the above numbers are final.	§5
7 — Repo cleanup	Resolve/delete files per §10.1, adopt the §10.2 structure, verify from a clean checkout.	§10
8 — Writing	Draft using the research document's full outline (§12), with every figure from Phase 6 and every table from Phases 1–5 already final.	Research doc §12

Do not start writing until Phases 0–5 are done. Every one of your 10 requested changes lands in a number, a figure, or a table that the paper needs — writing around numbers that are still going to change is the single most common reason a first draft has to be substantially reworked.